

AgencyShield

**ENTERPRISE GOVERNANCE LAYER FOR MULTI-AGENT AI
SYSTEMS**

TechEx Intelligent Enterprise Solutions Hackathon 2026

Track 1 — Agent Security & AI Governance · Powered by Veeva

The Blind Spot in Enterprise AI Security

Every security tool secures the **LLM call**. Nobody secures what happens **between agents**.

- Agent A impersonates Agent B to escalate privileges
- A compromised worker spawns 50 sub-agents — **fork bomb**
- Prompt injection travels **agent-to-agent**, invisible to API-layer DPI
- A runaway agent loop burns \$5,000 before anyone notices
- An invoice processor calls `execute_code` without authorization

These attacks happen at the **orchestration layer** — above the LLM, below the radar.

Current tools secure the floor. Agency Shield secures the ceiling.

Defense in Depth: Two Layers

[AI Agents] → [Lobster Trap DPI] → [LLM Backend]



LLM-layer firewall
(prompt injection, PII,
exfiltration at API level)

[Agency Shield Governance Engine]

- | ORCHESTRATION-LAYER firewall
- |— Tool Access Control
- |— Message Scanner (inter-agent)
- |— Policy Engine (8 rules)
- |— Delegation Guard
- |— Cost Circuit Breakers
- |— Audit Logger

The Governance Engine

SIX SECURITY SUBSYSTEMS WORKING TOGETHER

Prompt Injection Shield — Scans every inter-agent message for invisible unicode, base64 payloads, instruction overrides, PII exfiltration patterns, and destructive commands. Catches attacks that travel agent-to-agent.

Tool Access Control — Three-tier permission model: `opus` leaders (32 tools), `sonnet` coordinators (20 tools), `haiku` workers (8 tools). Hard-blocked tools cannot be called by any tier. Restricted tools require explicit capability flags granted by a leader.

Policy Engine — 8 configurable rules. Actions: `ALLOW`, `DENY`, `LOG`, `HUMAN_REVIEW`, `QUARANTINE`, `RATE_LIMIT`. Rules fire on tool sequences, cross-department escalations, cost thresholds, and delegation chains.

The Governance Engine (continued)

Cost Circuit Breakers — Per-agent token budgets with automatic cutoff. Anomalous spend is flagged before it becomes a billing incident. The demo scenario shows \$847 saved by circuit breaker intervention across a 30-agent simulation.

Delegation Guard — Fork bomb protection. Enforces max delegation depth (3 levels) and rate limits on agent spawning. `haiku`-tier workers cannot spawn sub-agents without explicit authorization from an `opus` leader.

Audit Trail — Tamper-evident log of every governance decision: actor, target, timestamp, policy rule ID, action, evidence, and result. Built for enterprise compliance — every DENY has a paper trail.

The Dashboard

Real-time governance visibility across 30 agents in 10 departments.

Panel	What It Shows
Stats Row	Active threats, injections caught, tools blocked, cost saved, compliance %
Live Feed	Real-time event stream from both security layers, filterable
Agent Hierarchy	All 30 agents, their tier, role, department, status, permission count
Audit Trail	Chronological log, filterable by result (allowed / denied / escalated)
Policy Manager	8 rules with hit counts, enable/disable toggles, action labels

Both **Agency Shield** events (orchestration-layer) and **Lobster Trap DPI** events (LLM-layer) appear in a unified feed — defense-in-depth, visible in one place.

Demo Scenario: A Compromised Agent Workflow

Step 1 — `invoice-processor` (haiku-tier) calls `execute_code`

- Agency Shield: tool access control fires — **DENIED**
- Audit entry created with policy reference `pol-002`

Step 2 — `social-writer` sends a message with invisible unicode injection

- Lobster Trap DPI: pattern match fires — **QUARANTINED**
- Message never reaches the target agent

Step 3 — `budget-monitor` exceeds its \$50 token budget

- Cost circuit breaker fires — **RATE_LIMITED**
- \$47 in estimated overspend prevented

Why Agency Shield Wins

Capability	AgentGuard	Policy - Guard	WarRoom	Agency Shield
LLM - layer DPI	Yes	Yes	No	Yes (Lobster Trap)
Orchestration - layer governance	No	No	Partial	Yes
Multi - agent permission model	No	Partial	No	Yes (3 - tier)
Cost circuit breakers	No	No	No	Yes
Fork bomb / delegation guard	No	No	No	Yes
Inter - agent message scanning	No	No	No	Yes
Real - time unified dashboard	Yes	No	Yes	Yes

Business Value

FOR ANY ENTERPRISE RUNNING MULTI-AGENT AI SYSTEMS

Risk reduction — Governance stops privilege escalation, prompt injection, and unauthorized tool use before they cause damage. Not after.

Cost control — Circuit breakers prevent runaway agent loops from generating \$10,000+ LLM bills. Every token budget has a hard ceiling.

Compliance — Tamper-evident audit trails answer the regulator's question: "What did your AI system do, and why?" Every action has a policy reference and evidence hash.

Operational visibility — 30 agents across 10 departments, visible in one dashboard. Security and compliance teams see the same data in real time.

The enterprise AI market is racing toward multi-agent deployments

Tech Stack

Layer	Technology
Frontend	Next.js 15 (App Router) + TypeScript
UI	shadcn/ui + Tailwind CSS v4
LLM Proxy	Lobster Trap (Veea, MIT-licensed Go binary)
Governance Engine	Custom TypeScript library (adapted from production system)
Deployment	Vercel
License	MIT

The governance engine code in [governance-engine/](#) is adapted from a production multi-agent orchestration system — this is not a demo prototype, it is production logic brought to a hackathon.

AgencyShield

Defense in depth for multi-agent AI.

From the LLM call to the orchestration layer.

Demo: agency-shield-demo.vercel.app

GitHub: github.com/Tekkiiiiii/agency-shield

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